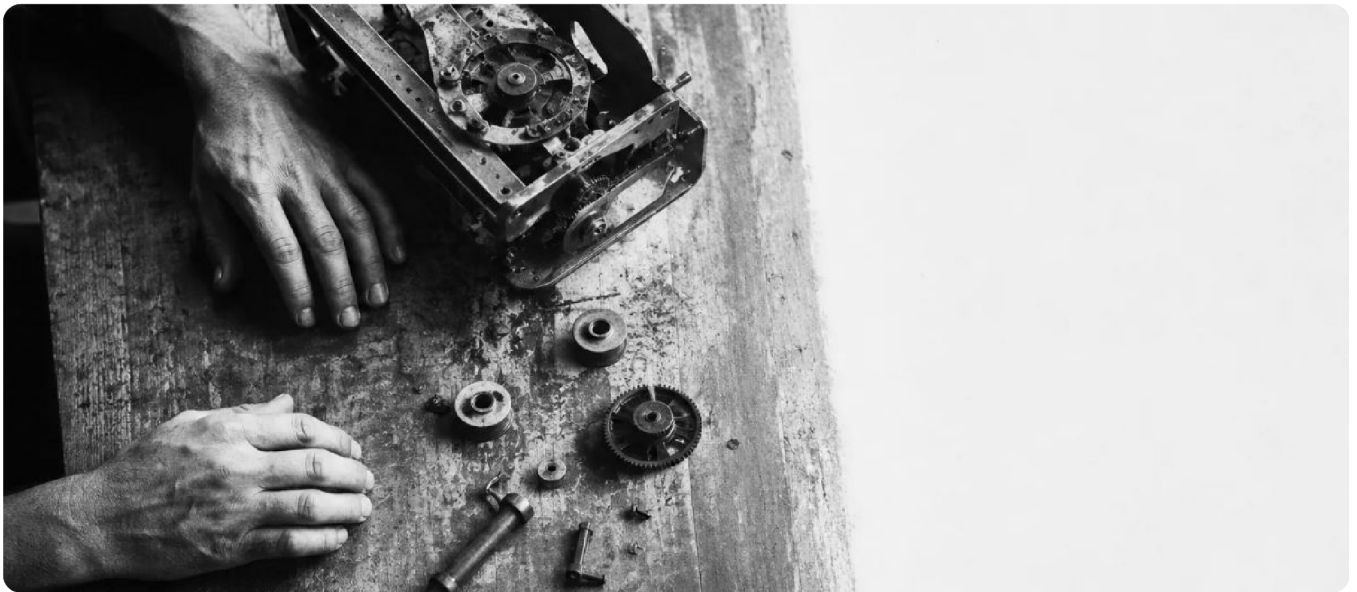


Work is being taken apart and put back together. Not thrown away.

What the research actually establishes about AI and work, what it projects to 2030, and why nobody credible forecasts 2045.



01

Augmentation exceeds automation

Assistance, not replacement is the larger category, in every income group

◦ ILO 2023

02

1 in 4

One worker in four is in a job with any measurable AI exposure

◦ ILO 2025

03

+78M

Net jobs still rising on the most-cited projection to 2030

◦ WEF 2025

04

2034

Nothing reaches 2045 the furthest credible projection stops at 2034

■ BLS 2025

Every figure in this publication carries its source and one of four labels. Where sources disagree, both are shown.

How the basis labels work

Every figure in AETHER carries one of these four, on screen and in print. Nothing is quoted without one.

■ MEASURED

Observed data, already collected and published by the cited source.

■ SCENARIO

A conditional pathway. Not a forecast.

◦ PROJECTION

A modelled forward estimate published by the cited source, sensitive to its assumptions.

◻ DESIGN ASSUMPTION

Our own assumption for this prototype, carried by no external source.

Photographs in this publication are illustrations made for it, not records of real workplaces. Each is captioned with the sourced figure it stands beside, so no image is ever doing the work of evidence.

8

DECENT WORK AND
ECONOMIC GROWTH

Aligned with Sustainable Development Goal 8

Goal 8 asks for sustained, inclusive and sustainable economic growth, full and productive employment, and decent work for all. Both halves matter here: this report argues about productivity as much as about jobs. Five targets are where it lands, and each is a place the evidence shows a gap rather than progress.

8.2 Higher levels of economic productivity through technological upgrading and innovation

Indicator 8.2.1 is output per worker, and that is exactly the quantity credible estimates disagree about by an order of magnitude. Section 08.

8.5 Full and productive employment and decent work for all women and men, and equal pay for work of equal value

Indicator 8.5.2 is the unemployment rate. Exposure also falls unevenly by sex: female-dominated occupations are almost twice as exposed. Section 07.

8.8 Protect labour rights and promote safe and secure working environments for all workers

Indicator 8.8.2 measures labour-rights compliance. Algorithmic management is spreading beyond platform work, and formal oversight can be substantively hollow. Section 06.

8.3 Encourage the formalization and growth of enterprises

Indicator 8.3.1 is the informal share of employment. 2.1 billion people work informally and sit outside every projection in this report. Sections 01 and 10.

8.6 Substantially reduce the proportion of youth NOT in employment, education or training

The only target here with a deadline already passed, 2020, and the share has since risen. The ILO flags AI as a risk factor for educated first-job seekers, and 11 in every 100 workers are projected to need training and not receive it. Sections 01 and 10.

Twelve sections

Each opens with its finding in one line. The full analysis sits behind a toggle, so the depth is there if you want it and out of the way if you do not.

01 **What the evidence says**

The four findings, in one screen

03 **Which skills are moving**

Rising, declining, and the deceleration

05 **Replaced, or assisted**

The ILO revised its own estimate down

07 **Who gets left out**

Access comes before skills

09 **Three ways 2045 could go**

Conditions, not predictions

11 **What it comes to**

A narrower claim than either side makes

02 **How far ahead we can see**

Why 2045 is scenario space

04 **Where jobs actually grow**

Two lists that barely overlap

06 **What protects a worker**

One binding law, two recommendations

08 **What it does to the economy**

Credible estimates disagree tenfold

10 **What should be done**

Six, each tied to an evidence gap

12 **Where it all came from**

23 sources, and what each cannot show

What the evidence says

Four findings hold across the sources reviewed here.

- Tasks are being reallocated inside jobs faster than jobs are being eliminated. That is well evidenced. What happens after 2030 is not.

24%

Share of clerical tasks that are highly exposed

◦ ILO 2023

4.9%

Projected global unemployment rate for 2026

◦ ILO 2026

22%

Total labour-market churn as a share of current formal jobs

◦ WEF 2025

39%

Share of workers' core skill sets employers expect to change by 2030

◦ WEF 2025



-
- 24%** ◦ ILO 2023 Exposure is concentrated rather than general. Clerical work is the only broad occupational group where a substantial share of tasks is highly exposed.



2.1B ◦ ILO 2026 And the surveys behind these projections see formal employment. Informal work, around two billion people, is largely invisible to them.

Four findings hold across the sources reviewed here. First, exposure to generative AI is real but concentrated: around one in four workers globally is in an occupation with some exposure, and 24% of clerical tasks are highly exposed against 1% to 4% for every other broad occupational group ILO 2023. Second, augmentation potential exceeds automation potential in every income group ILO 2023, which inverts the assumption most public discussion begins from.

Third, aggregate employment has not collapsed and is not projected to. Unemployment is projected stable near 4.9% ILO 2026, and the most-cited employer projection for 2030 is net growth of 78 million jobs WEF 2025. The disruption appears as churn, equal to 22% of formal jobs, and as pressure on job quality rather than on headline unemployment.

Fourth, and most consequentially for a product about 2045: **no institutional projection in this evidence base reaches past 2034** BLS 2025. The ILO describes its own exposure classifications as only illustrative and as a static snapshot ILO 2025, and states that the outcomes of the transition are not predetermined because humans decide how to incorporate these technologies ILO 2025.

The useful question is not whether a job survives, but which of its tasks change and what its holder learns next.

WHAT THIS SUMMARY DOES NOT CLAIM

That AI is harmless. Exposure is concentrated in clerical work, which is disproportionately performed by women: female-dominated occupations are almost twice as exposed as male-dominated ones [ILO 2026](#). Roughly 11 in every 100 workers are projected to need training and not receive it [WEF 2025](#), which converts to over 120 million people.

That the aggregate figures describe everyone. The employer surveys behind most projections cover 1.2 billion formal jobs. Around 2.1 billion people work informally and are largely invisible to this evidence base [ILO 2026](#).

02

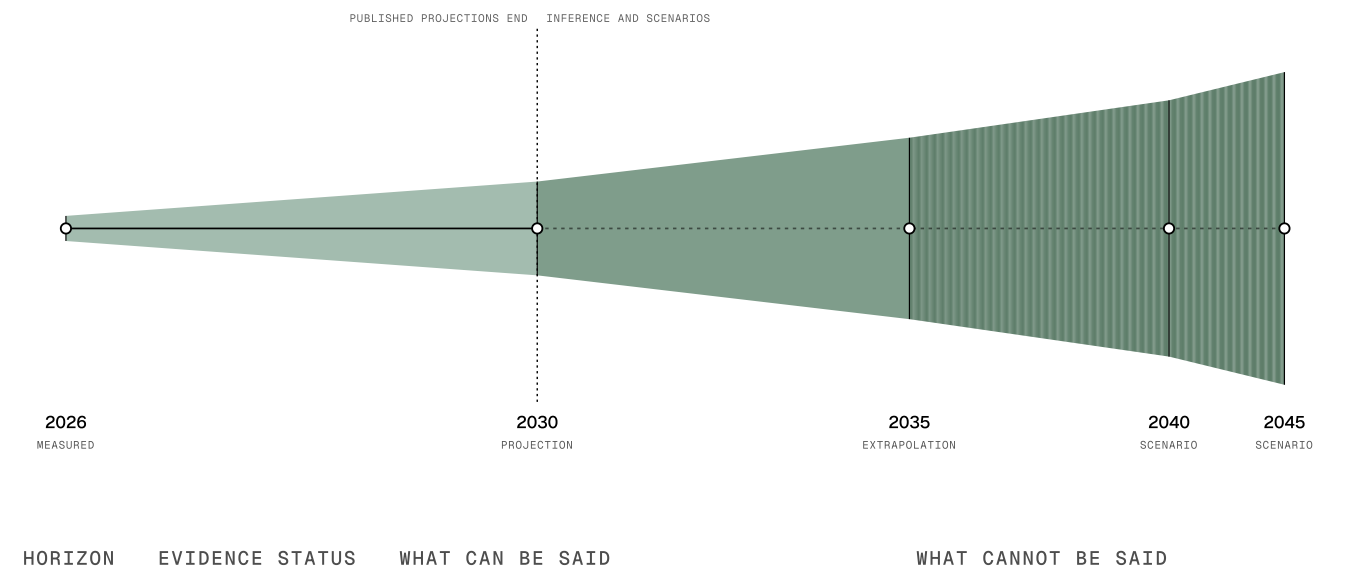
How far ahead we can see

Certainty degrades with distance.

— Published projections stop at 2034. Past that, this publication changes the register of its claims rather than extending the line.

Nobody can see 2045. The honest thing is to show where sight ends.

The band is what can defensibly be said at each horizon. It widens because the range genuinely widens, and turns to texture where published evidence runs out and inference begins.



HORIZON	EVIDENCE STATUS	WHAT CAN BE SAID	WHAT CANNOT BE SAID
2026	Measured	One in four workers has some generative-AI exposure. Unemployment is stable near 4.9%. Exposure concentrates in clerical work.	That any measured job losses are down to AI. No source measures this.
2030	Projection	Employers expect 170 million jobs created and 92 million displaced, a net gain of 78 million, and 39% of core skills to change.	That these figures will occur. They are survey extrapolations across 1.2 billion formal jobs.
2035	Extrapolation	On current direction of travel, task reallocation within occupations would continue to outpace whole-occupation elimination.	Any figure at all. No published projection reaches 2035. This horizon is our inference.
2040	Scenario	Which pathway is unfolding could be identified from leading indicators observable before 2035.	Which pathway will occur, or that any named occupation will or will not exist.
2045	Scenario	Three conditional pathways can be described, with the conditions each requires stated explicitly.	Anything quantitative. Outcomes depend on deployment and policy choices not yet made.

MEASURED

Data already collected and published.

PROJECTION

A source's own forward estimate, to 2030.

EXTRAPOLATION

Ours, from direction of travel. No source's authority.

SCENARIO

Conditional pathways. Not forecasts.

Hover or focus any horizon for what can and cannot be said at that point. The band's vertical extent encodes how wide the range of defensible statements becomes; it is an illustration of uncertainty, not a measured confidence interval.

The four registers are not stylistic. At 2026 the sources report what they measured. At 2030 they report what they model, and the WEF figures in particular are extrapolations from employer expectations rather than econometric forecasts. At 2035 nothing published reaches that far, so any statement is our own inference from direction of travel and carries no source's authority. At 2040 and beyond only conditional pathways are defensible.

Two asymmetries run through every horizon and are permanent features of this evidence base rather than caveats to be noted once. The datasets see the formal-sector minority. And exposure is not job loss: it measures task overlap with model capability, which is the most commonly misread quantity in this field IMF 2024.

Five dimensions of transformation, traced across the horizons. The verb register changes with the evidence: is, would, could.

DIMENSION	MEASURED NOW	EXTRAPOLATED	SCENARIO SPACE
Automation	Clerical work: 24% of clerical tasks highly exposed, against 1-4% elsewhere.	Would extend into digitised professional and technical roles.	Could reach context-dependent judgement, or stall there. The ILO offers its estimates as a static snapshot, not a trajectory.
Augmentation	Larger in every income group: 10.4% to 13.4% potential against 0.4% to 5.5% for automation.	Would stay the dominant effect.	Could become the organising principle of work, if integration methods are found.
Human-AI collaboration	Combinations beat humans alone, $g = +0.64$, but underperform the better of the two, $g = -0.23$.	Depends on solving the synergy problem. Currently unsolved.	Could deliver broad gains, or stay confined to creative work if the decision penalty holds.
Skill demand	AI and big data fastest-growing. Analytical thinking still the top core skill.	Would shift further toward judgement about when to rely on a system.	No skill can be named as valuable in 2045. The boundary moves with capability.
Work quality	Algorithmic management spreading beyond platforms. 2.1 billion in informal work. Stability called fragile.	No published projection exists. Measured at 2026 and nowhere else.	Could improve with real oversight, or worsen under nominal compliance. Governance decides, not technology.

Which skills are moving

Rising, declining, and a deceleration nobody expected.

— Employers expect 39% of core skills to change by 2030. That figure has fallen from 57% in 2020, which is the more interesting finding.

39%

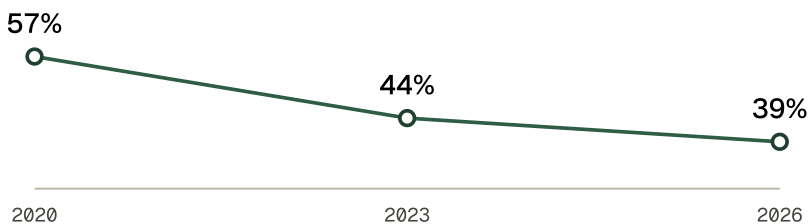
Share of workers' core skill sets employers expect to change by 2030

Down from 44% in 2023 and 57% in 2020. The deceleration is part of the finding. Skill sets, not jobs.

◦ PROJECTION · WEF 2025

Skill disruption is decelerating, not accelerating

Share of core skills employers expect to be transformed or outdated within five years, as reported in three successive editions of the same survey.



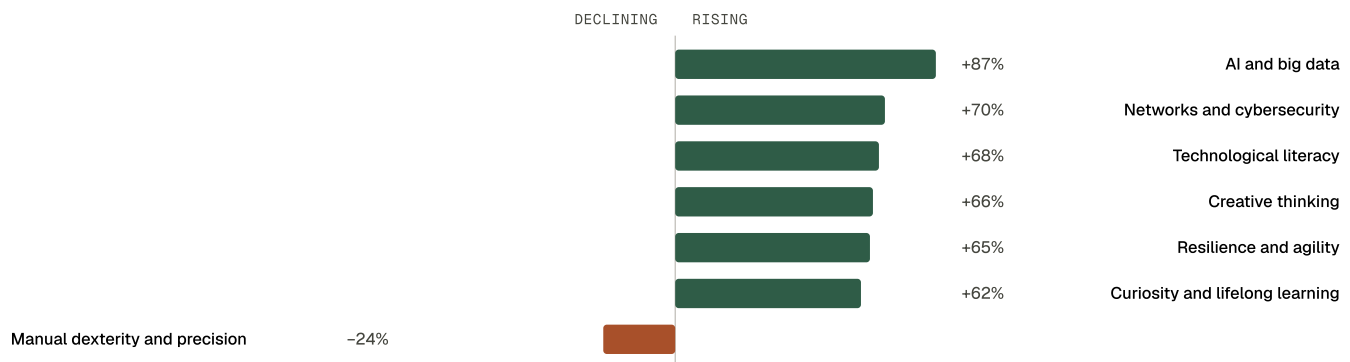
If AI capability were driving skill disruption upward, this series would rise. Three points are a direction, not a trend.

◦ WEF 2025 · Share of workers' core skill sets employers expect to change by 2030. World Economic Forum, 2025.

Fastest growing is not the same as most important. AI and big data lead the growth ranking WEF 2025, but analytical thinking remains the top core skill employers name, and what they describe wanting is technical and human skills together rather than one substituting for the other WEF 2025.

Which skills are employers adding, and which are they losing?

Net share of employers reporting each skill rising or declining in importance. Side of the axis carries the direction; colour is secondary.



SKILL	NET CHANGE	DIRECTION
AI and big data	+87%	Rising
Networks and cybersecurity	+70%	Rising
Technological literacy	+68%	Rising
Creative thinking	+66%	Rising
Resilience and agility	+65%	Rising
Curiosity and lifelong learning	+62%	Rising
Manual dexterity and precision	-24%	Declining

- WEF 2025 The fastest-growing skill to 2030. World Economic Forum, 2025.
- WEF 2025 Net share of employers reporting manual dexterity and precision in decline. World Economic Forum, 2025.

59 in 100

Workers needing training by 2030

- WEF 2025

1/3 vs 1/2

Training participation, workers aged 60-65 against those aged 25-44

- OECD 2025

63%

Employers naming skills gaps as the top barrier to transformation

- WEF 2025

WHAT THIS DOES NOT SAY

The 39% is a share of skill sets, not of jobs. No measurement of AI literacy exists for any population, so nothing here supports a claim about population-level attainment. And no specific skill can be named as valuable in 2045: the boundary between complemented and substituted tasks moves

with capability, and the ILO presents its own estimates of that boundary as a static snapshot

ILO 2025.

04 Where jobs actually grow

Two lists of growing occupations that barely overlap.

- Ranked by percentage, technology roles lead. Ranked by number of jobs, care and frontline work leads. Conflating the two is the standard error in careers advice.

Ranked by percentage growth, technology roles lead: Big Data Specialists, FinTech Engineers, AI and Machine Learning Specialists, Software Developers WEF 2025. Ranked by absolute growth, frontline and care roles lead, and the driver is demography rather than AI WEF 2025.

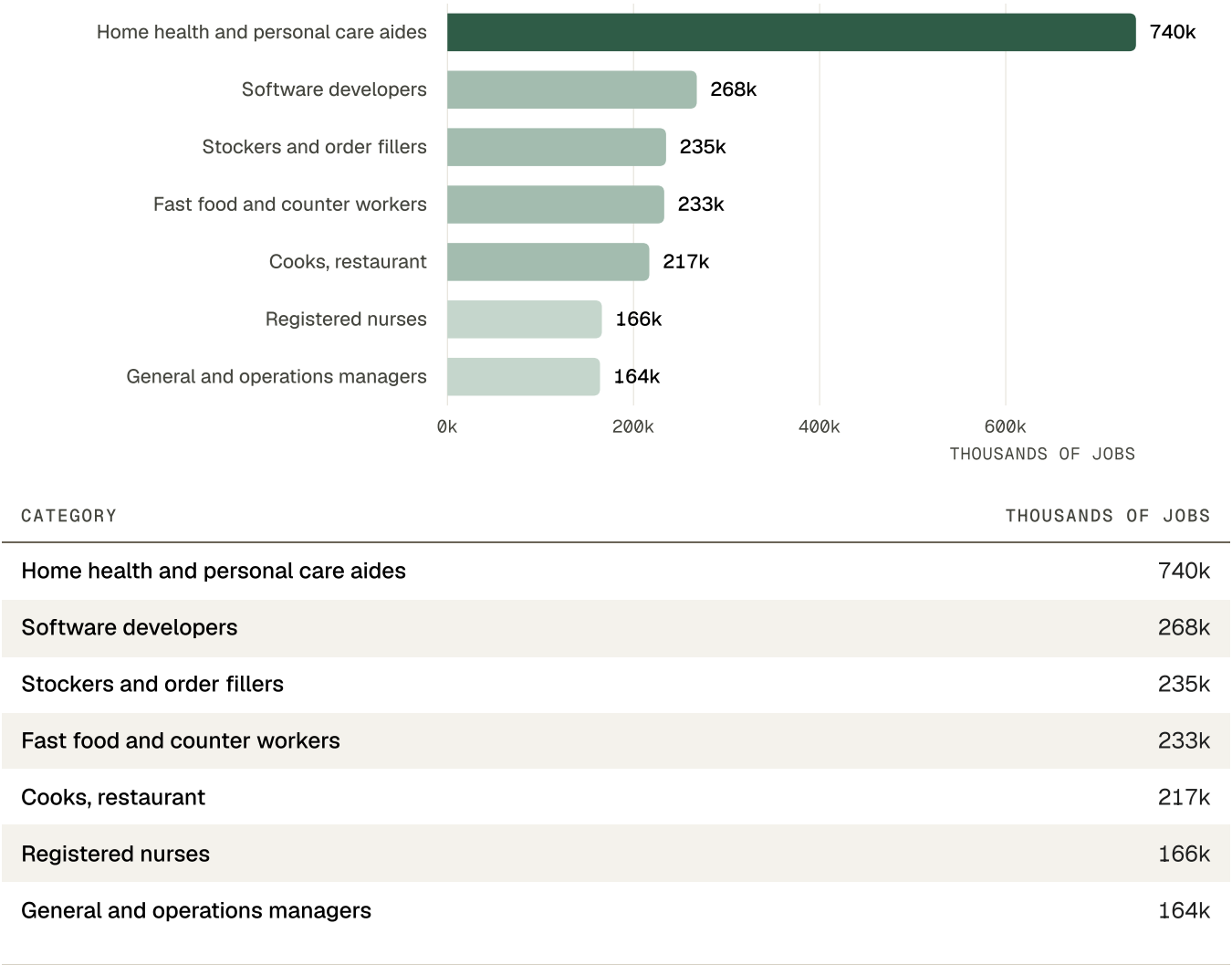
The distinction is not academic. The two fastest-growing occupations in the US projections add fewer than 20,000 jobs combined, a caveat the Bureau of Labor Statistics supplies itself BLS 2025. Meanwhile home health and personal care aides add 739,800 positions at a median wage of \$34,900, against software developers at 267,700 and \$133,080 BLS 2025.

Growth in jobs is not growth in good jobs, and percentage growth is not scale.

Where is US employment actually growing in absolute terms?

Projected change in jobs, 2024 to 2034. Bar length is the number of new positions.

Median pay is in the data table, because growth and quality are different questions.



- BLS 2025 Largest absolute US growth: home health and personal care aides. US Bureau of Labor Statistics, 2025.
- BLS 2025 Fastest-growing US occupation: wind turbine service technicians. US Bureau of Labor Statistics, 2025.



+739,800 ◦ BLS 2025 Ranked by number of jobs, growth looks like this: the largest single source of new US employment, at a median wage of \$34,900.



+49.9% ◦ BLS 2025 Ranked by percentage, it looks like this: the fastest-growing US occupation, from a base of 13,600 jobs. Both lists are true and they answer different questions.

The six career categories this project's dashboard presents, assessed against the evidence. Only one has direct projection support.

CAREER	STATUS	NEAREST EVIDENCED ROLE	HONEST READING
Climate Technology Specialist	Growing	Renewable Energy Engineers; wind turbine technicians +49.9%	The best evidenced of the six, in both WEF and BLS. Show absolute numbers beside the percentage or the small base misleads.
AI Safety Specialist	Emerging	Information Security Analysts +29%	Strongest regulatory anchor of the six: the AI Act requires oversight by competent, trained, authorised persons. A legal mandate, but no projection counts these jobs.
AI Governance Specialist	Emerging	Security Management Specialists	Rests on the same anchors as AI Safety Specialist. The evidence does not separate the

CAREER	STATUS	NEAREST EVIDENCED ROLE	HONEST READING
			two, so they should be merged or explicitly distinguished.
Human-AI Interaction Designer	Emerging	UI and UX Designers	Best mechanistic case: interaction choices determine whether teaming gains or loses. But the standard levers, explanations and confidence displays, were insignificant across 370+ effect sizes.
Robotics Systems Engineer	Emerging	Autonomous and Electric Vehicle Specialists	58% of employers expect robotics to transform their business. Countervailing: physical dexterity has repeatedly resisted automation.
AI Product Designer	Emerging	UI and UX Designers	Weakest of the six. Graphic designers are newly on the fastest-declining list with generative AI named as the cause, so design is not uniformly a growth area.

05

Replaced, or assisted

The distinction the ILO stopped treating as binary.

— After two years of real-world use the ILO revised its own peak automation estimate down, from 0.9 to 0.76. Assistance is the larger category in every income group.

0.9 → 0.76

Peak task-level automation score, 2023 estimate against 2025 estimate

Two years of real-world use lowered the ILO's own peak estimate. Highest occupational mean is 0.7.

■ MEASURED · ILO 2025

In 2023 the ILO's highest task-level automation score reached 0.9. In the 2025 refined index the highest is 0.76, with the highest occupational mean at 0.7 ILO 2025. Two years of real-world deployment caused the source to revise its own estimate downward. A source correcting itself after contact with reality is stronger evidence than either estimate alone.

The same update replaced the binary automation-or-augmentation split with a four-gradient spectrum ILO 2025. This has a direct design consequence for any product built on it: exposure should be rendered as gradient position with task-level variance, never as a safe-or-at-risk verdict.

How is global employment distributed across the exposure gradients?

Segments are an ordered sequence from no measurable exposure to the highest gradient, so the ramp runs light to dark. A 2px gap separates adjacent fills.



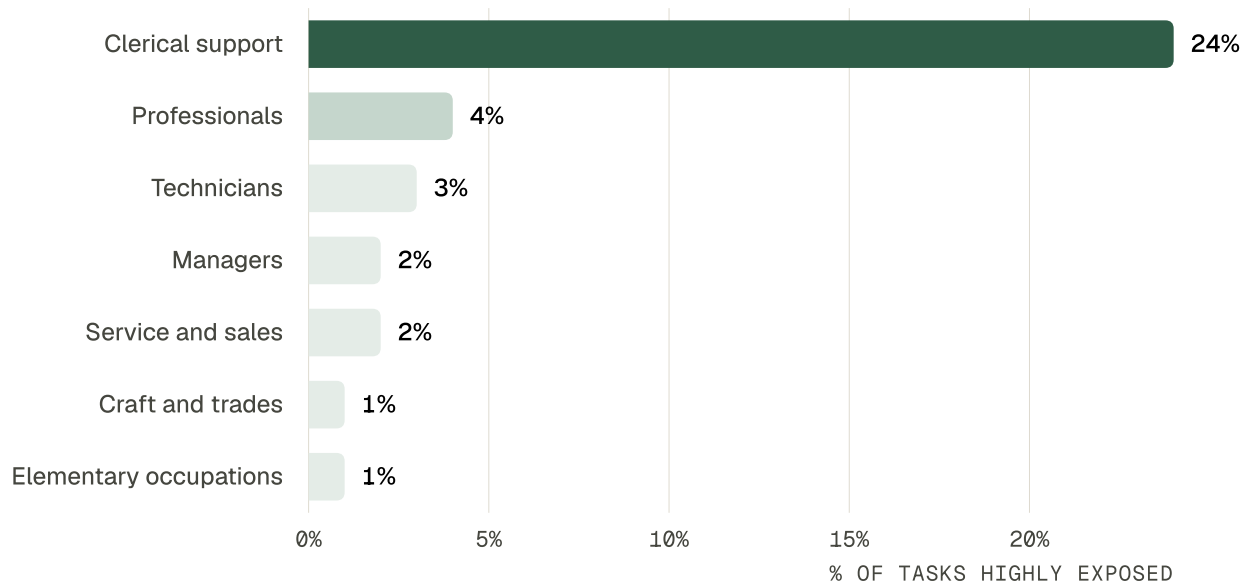
SEGMENT	SHARE (%)
No measurable exposure	75%
Gradient 1-2	15%
Gradient 3	6.7%
Gradient 4	3.3%

Gradient 4 is 3.3% of global employment, up from 2.3% in the previous edition despite lower task-level scores. Both movements are shown, because reporting either alone would mislead.

- ILO 2025 Workers globally in an occupation with some generative-AI exposure. International Labour Organization, 2025.
- ILO 2025 Share of employment in the highest exposure gradient, between ILO editions. International Labour Organization, 2025.

Is exposure evenly spread across occupations?

Share of tasks rated highly exposed, by broad occupational group. The concentration is the finding.



CATEGORY	% OF TASKS HIGHLY EXPOSED
Clerical support	24%
Professionals	4%
Technicians	3%
Managers	2%
Service and sales	2%
Craft and trades	1%
Elementary occupations	1%

o ILO 2023 Share of clerical tasks that are highly exposed. International Labour Organization, 2023.

WHERE THIS READING IS CONTESTED

Two findings run against the augmentation-dominant reading and are stated rather than omitted. The share of employment in the highest gradient rose from 2.3% to 3.3% between editions ILO 2025, and exposure expanded into strongly digitised professional and technical roles ILO 2025. Every estimate here was also computed against 2023 to 2025 model generations.

What protects a worker

One binding instrument, two recommendations, and a documented failure.

- The EU AI Act makes workplace AI high-risk and requires a human who can override it. EU guidance also documents a case where that oversight existed on paper while the algorithm decided.

What the law requires, and what guidance found in practice

The EU AI Act is the only binding instrument in this evidence base. European Commission guidance documents what its requirements look like when they are met formally rather than substantively.

REQUIRED

EU AI Act

- Employment AI classified high-risk (Annex III)
- Human oversight with genuine override and stop (Art. 14)
- Automation bias named explicitly
- Workers and representatives informed before deployment (Art. 26)
- Oversight by persons with competence, training and authority

THE GAP

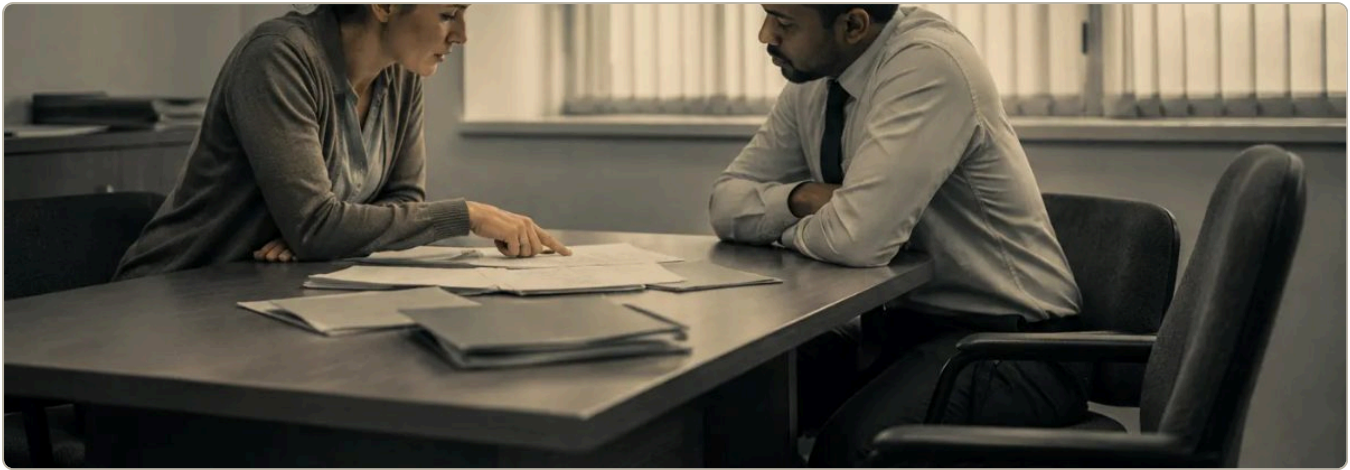
OBSERVED

A documented pay-setting case

- Oversight existed formally
- The algorithm effectively decided
- Appeals rarely changed the outcome

- EU 2024 How the EU AI Act classifies employment and worker-management AI. European Union, 2024.
- EU 2024 Human oversight will have to include genuine override and stop capability. European Union, 2024.
- EU 2024 Employers will have to inform workers and representatives before deploying high-risk AI. European Union, 2024.
- European Commission 2026 Formal human oversight can exist while the algorithm effectively decides. European Commission, 2026.

Compliance and control are not the same thing.



Oversight can be hollow ■ European Commission 2026 What the requirement looks like when it is met formally rather than substantively: a review happens, and the outcome rarely changes.

Governance instruments covering AI at work, with what each actually requires and its force.

INSTRUMENT	FORCE	REQUIRES	LIMIT
EU AI Act (2024/1689)	Binding law	High-risk classification for employment AI. Human oversight with override. Pre-deployment worker notification. Log retention.	EU-linked deployment only, in phased application.
UNESCO Recommendation	Non-binding, all member states	Redress mechanisms, auditability, prohibition where use would violate human rights. Suggests an AI Ethics Officer role.	No enforcement mechanism.
OECD AI Principles	Non-binding, 47 adherents	Human agency and oversight safeguards. Labour rights named among risks requiring systematic management.	A recommendation.
ILO standards	No AI-specific instrument verified	Research and calls for social dialogue on algorithmic management.	No Convention or Recommendation specifically governing AI at work was found. Do not imply one exists.

The structural problem beneath the specific risks is information asymmetry. Algorithmic management concentrates knowledge and control on the employer's side, so workers are managed by a system whose logic they cannot

see [Baiocco, Fernandez-Macias, Rani & Pesole 2022](#). It is spreading beyond platform work into customer service, transport, logistics, banking and healthcare, and the ILO notes it does not require AI to operate [ILO 2024](#).

WHAT CANNOT BE QUANTIFIED HERE

No quantified rate of AI hiring bias exists in this evidence base. The Amazon recruiting case is a single documented instance from 2018, reported rather than studied, and is cited as illustration only [Reuters 2018](#). No surveillance-prevalence figure exists for the general workforce: the 40% tracking figure covers surveyed freelance platform workers specifically

[Baiocco, Fernandez-Macias, Rani & Pesole 2022](#).

Job quality and algorithmic management are measured at 2026 and nowhere else. There is no projection to extend, so this section cannot show a forward trajectory for its own subject matter.

07

Who gets left out

Access comes before skills.

— 2.2 billion people are offline. Every reskilling proposal assumes they are not, and training participation collapses with age.

2.2B

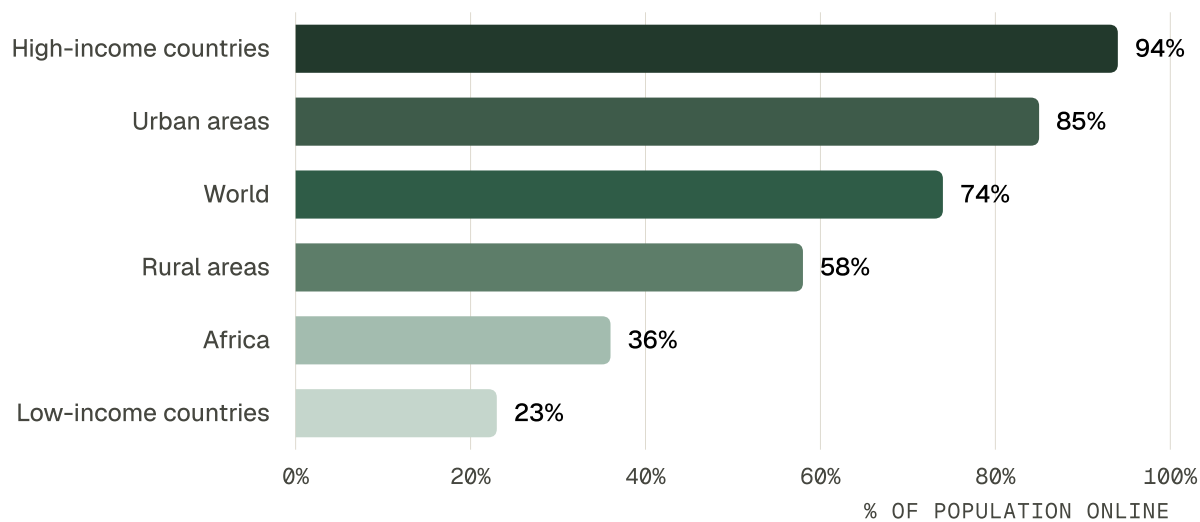
People still offline

6.0 billion online, 74% of the world. Reskilling assumes access these people do not have.

■ MEASURED · ITU 2025

Who has the connectivity that reskilling assumes?

Share of population using the internet, by group. This is the floor beneath any digital-skills claim.

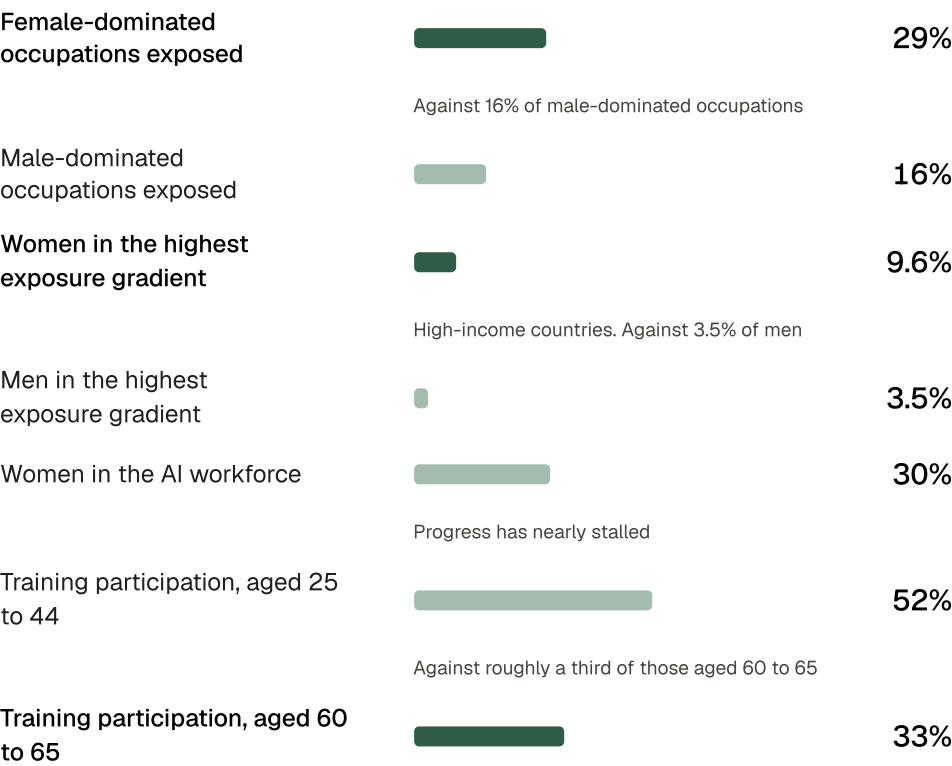


CATEGORY	% OF POPULATION ONLINE
High-income countries	94%
Urban areas	85%
World	74%
Rural areas	58%
Africa	36%
Low-income countries	23%

- ITU 2025 Internet use in high-income against low-income countries. International Telecommunication Union, 2025.
- ITU 2025 Urban against rural internet use. International Telecommunication Union, 2025.
- ITU 2025 Internet use in Africa. International Telecommunication Union, 2025.

Who carries the exposure, and who gets the training

Each pair compares the same measure across two groups. The mechanism behind the gender gap is occupational segregation, not capability.



- ILO 2026 Exposure of female-dominated against male-dominated occupations. International Labour Organization, 2026.
- ILO 2025 Female against male employment in the highest exposure gradient, high-income countries. International Labour Organization, 2025.
- ILO 2026 Women's share of the AI workforce. International Labour Organization, 2026.
- OECD 2025 Training participation, workers aged 60-65 against those aged 25-44. OECD, 2025.



29% vs 16% ◦ ILO 2026 Occupational segregation is the mechanism, not capability: the most exposed group of tasks is also the one most performed by women.



2.2B ■ ITU 2025 Before any of that, connectivity. A quarter of the world is not online, and every training proposal in this report assumes otherwise.

This is why aggregate training figures are the wrong measure. Completion can rise while exclusion persists, so reach among currently-excluded groups is the honest metric OECD 2025.

The people whose work is most exposed to these systems are underrepresented among the people building them.

What it does to the economy

Credible estimates disagree by an order of magnitude.

- The most rigorous sceptical estimate puts AI's ten-year productivity gain at 0.66%. Ageing already measured twice that between 2000 and 2019, in the opposite direction.

Two credible estimates, measuring different things

Both are sound. One models the whole economy over a decade; the other measures one occupation in one firm. Neither refutes the other, and quoting either alone misleads.

ECONOMY-WIDE, MODELLED

0.66% over ten years

- 0.66% total factor productivity, framed by its author as an upper bound
- 0.53% once hard-to-learn tasks are counted
- GDP effect 0.93% to 1.16%, or 1.4% to 1.56% if investment responds
- Rests partly on assumptions the author labels speculative

DIFFERENT SCOPES

ONE OCCUPATION, MEASURED

+15%, and +30% for the least experienced

- 5,172 customer support agents, one firm
- Small significant declines in quality for the most skilled
- A pre-2024 model and a single tool
- Strong identification, narrow scope

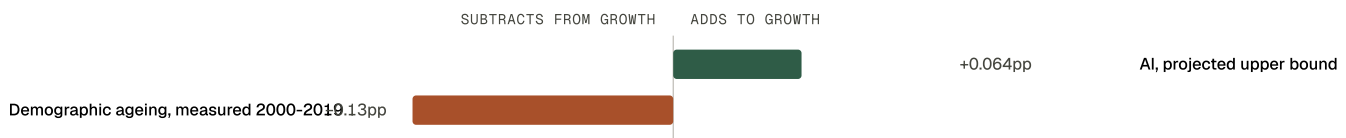
◦ Daron Acemoglu 2025 Estimated ten-year total factor productivity gain from AI. Daron Acemoglu, 2025.

◦ Daron Acemoglu 2025 The same estimate, adjusted for hard-to-learn tasks. Daron Acemoglu, 2025.

■ Brynjolfsson et al. 2025 Productivity gain from generative AI: average, and for the least experienced workers. Brynjolfsson, Li & Raymond, 2025.

How large is the projected AI productivity effect, in context?

Annual percentage-point contribution to productivity growth. The demographic figure is measured; the AI figure is a projection, and its source's own upper bound.



SKILL	NET CHANGE	DIRECTION
AI, projected upper bound	+0.064pp	Adds to growth
Demographic ageing, measured 2000-2019	-0.13pp	Subtracts from growth

Two different quantities share one axis deliberately: the comparison of scale is the point. Ageing is a measured drag, AI a projected gain.

- Daron Acemoglu 2025 Estimated ten-year total factor productivity gain from AI. Daron Acemoglu, 2025.
- OECD 2025 Annual drag on OECD productivity growth from ageing, estimated over 2000-2019. OECD, 2025.



+15% / +30% ■ Brynjolfsson et al. 2025 The cleanest measurement in this field covers one occupation in one firm: 5,172 support agents, where the least experienced gained most and the most skilled saw small quality declines.

On distribution the reading is more consistent, and less comfortable. AI is not expected to reduce inequality on its own: no evidence of labour income inequality relief, a widening capital-labour gap, and small real wage declines projected for some groups Daron Acemoglu 2025. The IMF's position runs both

ways in a single sentence, which is itself informative IMF 2024. Cross-country divergence is the more consistent warning: economies facing less immediate disruption are also least ready to capture the benefits IMF 2024.

WHAT THIS DOES NOT SAY

The IMF states directly that AI's exact economic implications are challenging to predict, comparing the uncertainty to past general-purpose technologies such as electricity IMF 2024. The sceptical estimates exist in two live versions, 0.71% in the 2024 working paper and 0.66% in the 2025 published version, and must not be mixed in one chart. Consultancy projections widely circulated in this space were deliberately excluded.

09 Three ways 2045 could go

Conditional pathways, with the conditions each one requires.

- Assistance-led transformation is where the evidence points. Whether that turns out well depends on governance, not on the technology.

The three differ on two axes, chosen because the evidence identifies both as unresolved and consequential: where deployment lands in the task distribution, and whether governance and reskilling delivery are substantive or nominal. The method follows the ILO's own, mapping drivers onto job quantity, job quality and inequality ILO 2017.

01

Responsible AI Adoption

Adoption is deliberate and governed, and reskilling reaches the people currently excluded.

Best outcomes, weakest support for its central assumption

PRODUCTIVITY

Higher than Scenario 02 on welfare, plausibly lower on measured output

JOB TRANSFORMATION

Task reallocation dominant, whole-occupation displacement lower

NEW OCCUPATIONS

Concentrated in governance, oversight and assurance functions

RESKILLING DEMAND

Highest of the three, and continuous rather than one-time

INEQUALITY

Narrower divergence than the alternatives, but not reduced

WORKPLACE CONDITIONS

Better on transparency and recourse

VOICE AND RECOURSE

Override and appeal change outcomes, because compliance is enforced

LEGIBILITY OF DECISIONS

A worker can inspect the logic that ranked them

CONDITIONS THIS PATHWAY REQUIRES

High-risk rules spread beyond the EU. Oversight is substantive. Training reaches the 11 in 100 projected to miss out. Connectivity improves.

WHAT IT CANNOT SUPPORT WITH EVIDENCE

That reskilling at scale works. No outcome evaluation exists: no completion rates against earnings, no cost per transition. The largest gap here.

02

High Automation

Capability advances into hard-to-learn tasks, and deployment optimises for substitution.

Least supported by the most recent revision, but not refuted

PRODUCTIVITY

Higher measured output, weaker translation into welfare

JOB TRANSFORMATION

Whole-occupation displacement highest, concentrated in clerical work

NEW OCCUPATIONS

Fewer relative to displacement, more of contested social value

RESKILLING DEMAND

Highest in volume, lowest in delivery

INEQUALITY

Diverging most sharply, on capital-labour, cross-country and gender lines

WORKPLACE CONDITIONS

Worst of the three. Algorithmic management extends beyond platforms

VOICE AND RECOURSE

Weakest. Oversight exists on paper while the system effectively decides

LEGIBILITY OF DECISIONS

Lowest. A worker cannot contest what they cannot inspect

CONDITIONS THIS PATHWAY REQUIRES

Capability generalises to context-dependent tasks. Deployment targets the 33% that is high-exposure, low-complementarity. Governance stays gapped.

WHAT IT CANNOT SUPPORT WITH EVIDENCE

That capability generalises to hard tasks. Real-world use cut the ILO's peak scores from 0.9 to 0.76. But top-gradient share did rise, 2.3% to 3.3%.

OBSERVABLE BEFORE 2035

- High-risk AI requirements adopted in jurisdictions beyond the EU
- Enforcement actions or documented remediation, not compliance paperwork alone
- Override and appeal rates that actually change outcomes
- Training completion rising among the 60–65 cohort, not just in aggregate
- Publication of the first credible reskilling outcome evaluations

■ SCENARIO · ILO 2017

OBSERVABLE BEFORE 2035

- Skill instability rising back past 39%, breaking the 57 to 44 to 39 deceleration
- A second and third non-clerical occupation entering the fastest-declining list
- ILO's next revision moving peak task scores back up toward 0.9
- Gradient-4 employment share rising materially above 3.3%
- Employer downsizing intentions rising above 41% while reskilling intent falls from 77%

■ SCENARIO · ILO 2017

03

Human-AI Collaboration

Work is reorganised around task-level complementarity rather than substitution.

Best supported as the direction of travel

PRODUCTIVITY

Gains concentrated in creation work, not decision work

JOB TRANSFORMATION

Highest role redefinition, lowest role elimination

NEW OCCUPATIONS

Collaboration design, workflow integration, interaction design

RESKILLING DEMAND

High, and weighted toward judgement rather than tool operation

INEQUALITY

Genuinely contested. Within-firm compression evidenced, macro-level not

WORKPLACE CONDITIONS

Better only if governance is substantive. Not automatic

VOICE AND RECOURSE

Depends entirely on governance. The technology does not decide this

LEGIBILITY OF DECISIONS

Better where work is designed around complementarity, not substitution

CONDITIONS THIS PATHWAY REQUIRES

Deployment targets augmentation and the high-complementarity 27% of advanced-economy employment. Needs working human-AI integration.

WHAT IT CANNOT SUPPORT WITH EVIDENCE

That the synergy problem gets solved. The pooled effect is negative, $g = -0.23$, and the most-proposed fix has 3 studies. An open question.

- Replication of the synergy findings against post-2023 models, the most informative indicator
- Delegation studies moving the effect to a confidence interval excluding zero
- The +15% and +30% productivity findings replicating outside single-firm settings
- Interaction-design roles appearing by name in a WEF or BLS ranked list
- Whether the decision-task penalty persists

■ SCENARIO · ILO 2017

Scenario 03 is where the direction of travel points. Scenario 01 is what would have to be added for that direction to be worth arriving at.

Stating a preference rather than treating the three as equally likely, because the evidence does lean and false balance is its own form of dishonesty. The strongest single reason: augmentation potential exceeds automation potential in every income group ILO 2023, and the revision that mattered most went against the high-automation pathway, with the ILO lowering its own peak task scores after real-world experience ILO 2025.

But support for Scenario 03 as a *direction* says nothing about whether its outcomes are good. Its core mechanism, effective human-AI integration, is currently an evidence gap with a negative pooled synergy effect Vaccaro et al. 2024. And collaboration-centred work is fully compatible with algorithmic management ILO 2024, so its conditions outcome depends on governance rather than on the collaboration design itself.

10 What should be done

Six, each tied to a specific gap in the evidence.

- The largest gap is that nobody has evaluated whether reskilling works. Every proposed response to AI and work assumes it does.
-

01 Evaluate reskilling, do not just fund it

The largest single gap in this evidence base is that no rigorous outcome evaluation of reskilling exists: no completion rates against employment or earnings effects, no cost per successful transition. Every proposed response to AI and work assumes reskilling works. Nobody has shown it.

02 Measure training reach, not training volume

Aggregate completion has risen from 41% to 50% while participation among workers aged 60 to 65 remains around one third. Volume can rise while exclusion persists, so reach among currently-excluded groups is the honest metric.

03 Treat oversight as a competence question

From December 2027 the AI Act will require oversight by persons with competence, training and authority. Commission guidance already documents what happens without it: formal review while the algorithm decides. Audit whether override actually changes outcomes, not whether a review step exists.

04 Publish exposure as gradient, never as verdict

The ILO abandoned the binary automation split for a four-step gradient. Any tool presenting a worker with a safe-or-at-risk judgement is misrepresenting the underlying research.

05 Fund replication of human-AI collaboration research

The meta-analytic finding that combinations underperform the better of human or AI alone covers studies up to June 2023. Whether it holds for current model generations is the single most decision-relevant open question in this field.

06 Extend measurement to informal work

Around 2.1 billion people work informally and are largely invisible to the employer surveys that produce most projections. Conclusions drawn from 1.2 billion formal jobs should not be stated as conclusions about work.

11

What it comes to

A narrower claim than either optimists or pessimists usually make.

- Tasks are being reallocated faster than jobs are being eliminated, the effects are concentrated rather than general, and 2045 is unknowable. The narrowness is the useful part.

What the research establishes is that tasks are being reallocated faster than occupations are being eliminated, that augmentation is the larger category in every income group, and that the effects are concentrated rather than general. What it does not establish is anything about 2045. The furthest institutional projection reviewed here reaches 2034, and the ILO presents its own exposure estimates as a static snapshot rather than a trajectory.

That combination is not a failure of the research. It is the finding. The outcomes of this transition are not predetermined, because humans decide how to incorporate these technologies and need to guide the transition ILO 2025. Which means the question worth putting to a person deciding what to learn is not what the future holds, but which tasks in their work are changing and what judgement they are building.

There is a second reading of that. If the outcome is decided by people rather than by the technology, then the thing to measure is not only how much work gets done but whether the person doing it can still question a decision made about them. The evidence says that is where the pathways actually diverge: oversight can exist on paper while the system decides European Commission 2026, and a worker cannot contest what they cannot inspect ILO 2024. Productivity is the easier thing to count. Dignity is the thing that has to be designed in, and it is the part no projection will deliver on its own.



Information asymmetry ■ Baiocco, Fernandez-Macias, Rani & Pesole 2022

The part no projection delivers. A worker cannot contest a decision they cannot inspect, which makes voice something to design in rather than an outcome to wait for.

Every confident claim about the workforce of 2045 is either a scenario wearing a forecast's clothes, or an invention. What is not a scenario is the choice: whether the systems built between now and then leave the people inside them able to be heard.

12 Where it all came from

Every figure resolves to an entry below.

— 23 sources, chosen for institutional authority and retrieved directly. Each entry states what it is good for and where it stops.

Sources were selected for institutional authority and retrieved directly. Consultancy projections widely circulated in this field were deliberately excluded in favour of intergovernmental, statistical-office and peer-reviewed material. Each entry notes what the source is authoritative for and where it stops, because most misreadings in this field come from stretching a source past its own stated scope.

Figures are transcribed to the source's own precision. Where a source revised its own estimate between editions, the revision is recorded as a finding rather than silently updated. Where sources disagree, both are shown.

INTERNATIONAL LABOUR ORGANIZATION

Generative AI and jobs: A Refined Global Index of Occupational Exposure (Working Paper 140) (2025)

The methodological backbone for occupational exposure. Builds a graded exposure index rather than a binary automatable/safe split, and abandons the automation/augmentation split of WP96. Its own classifications are described as only illustrative, carrying a degree of subjectivity, and as a static snapshot. Exposure measures task overlap with model capability, not employment outcomes. Its worker survey is Polish (1,640 respondents) and asks about the respondent's industry, not their own job; managers and groups 8 and 9 are over-represented, professionals most under-represented.

<https://webapps.ilo.org/static/english/intserv/working-papers/wp140/index.html>
ml <<https://webapps.ilo.org/static/english/intserv/working-papers/wp140/index.html>>

INTERNATIONAL LABOUR ORGANIZATION

Research Brief: Generative AI and jobs, a 2025 update (2025)

Accessible summary of Working Paper 140. Cite the working paper for method.

https://www.ilo.org/sites/default/files/2025-05/Research%20brief_GenAI%202025%20Update.pdf <https://www.ilo.org/sites/default/files/2025-05/Research%20brief_GenAI%202025%20Update.pdf>

INTERNATIONAL LABOUR ORGANIZATION

Generative AI and jobs: A global analysis of potential effects on job quantity and quality (Working Paper 96) (2023)

The earlier global exposure analysis, and the source of the automation/augmentation split figures and the clerical task-exposure shares. Predates current model generations. WP140 replaced its binary automation/augmentation split with graded exposure and carries no equivalent split figures.

https://www.ilo.org/sites/default/files/2024-07/WP96_web.pdf <https://www.ilo.org/sites/default/files/2024-07/WP96_web.pdf>

INTERNATIONAL LABOUR ORGANIZATION

Employment and Social Trends 2026 (2026)

ILO flagship. Current global employment conditions, job quality, and informality. The authoritative baseline for the state of work as it is now.

<https://www.ilo.org/publications/flagship-reports/employment-and-social-trends-2026> <<https://www.ilo.org/publications/flagship-reports/employment-and-social-trends-2026>>

INTERNATIONAL LABOUR ORGANIZATION

Gen AI, occupational segregation and gender equality in the world of work (2026)

The gendered distribution of AI exposure, driven by occupational segregation rather than by technology alone.

https://www.ilo.org/sites/default/files/2026-03/Research%20Brief%20GenAI_final0403_0.pdf <https://www.ilo.org/sites/default/files/2026-03/Research%20Brief%20GenAI_final0403_0.pdf>

BAIOCCO, FERNANDEZ-MACIAS, RANI & PESOLE

The Algorithmic Management of work and its implications in different contexts (EU-ILO Background Paper No. 9) (2022)

Algorithmic management and workplace monitoring as a labour issue. June 2022, a JRC co-publication in the EU-ILO background paper series, not an ILO Employment Working Paper. Its tracking figure covers freelance platform workers only.

https://www.ilo.org/sites/default/files/wcmsp5/groups/public/%40ed_emp/documents/publication/wcms_849220.pdf <https://www.ilo.org/sites/default/files/wcmsp5/groups/public/%40ed_emp/documents/publication/wcms_849220.pdf>

INTERNATIONAL LABOUR ORGANIZATION

Algorithmic management in the workplace (2024)

ILO's framing of algorithmic management and calls for social dialogue. Not a binding standard: no ILO Convention specifically governing AI at work was verified.

<https://www.ilo.org/algorithmic-management-workplace> <<https://www.ilo.org/algorithmic-management-workplace>>

INTERNATIONAL LABOUR ORGANIZATION

The Future of Work: A Scenario Analysis (2017)

Precedent for how the ILO itself handles long-horizon uncertainty: scenario sets, not forecasts.

<https://www.ilo.org/resource/future-work-scenario-analysis-0> <<https://www.ilo.org/resource/future-work-scenario-analysis-0>>

INTERNATIONAL LABOUR ORGANIZATION

Using foresight to think and act upon an uncertain future world of work: Trade unions' experiences (Working Paper 118) (2024)

The discipline of labour-market foresight, and why point predictions about the distant future are not credible.

https://www.ilo.org/sites/default/files/2024-07/118_web.pdf <https://www.ilo.org/sites/default/files/2024-07/118_web.pdf>

WORLD ECONOMIC FORUM

The Future of Jobs Report 2025 (2025)

The most-cited source for employer expectations on jobs and skills to 2030. Figures are extrapolations from a survey of just over 1,000 employers across 55 economies, scaled against 1.2 billion formal jobs. Not a census and not an econometric forecast; informal employment is largely out of scope.

https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf <https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf>

INTERNATIONAL MONETARY FUND

Gen-AI: Artificial Intelligence and the Future of Work (Staff Discussion Note SDN/2024/001) (2024)

Cross-country AI exposure, including the advanced versus emerging economy split. The widely-quoted 40% figure is occupational exposure, not predicted job loss. Staff views, not official IMF position.

<https://www.imf.org/-/media/files/publications/sdn/2024/english/sdnea2024001.pdf> <<https://www.imf.org/-/media/files/publications/sdn/2024/english/sdnea2024001.pdf>>

OECD

OECD Employment Outlook 2025: Can We Get Through the Demographic Crunch? (2025)

OECD flagship labour analysis, this edition on demographic ageing and labour supply. Contains no AI chapter: its demographic estimates are historical, covering 2000-2019. OECD-country focused; does not generalise to low-income economies.

https://www.oecd.org/en/publications/2025/07/oecd-employment-outlook-2025_5345f034.html <https://www.oecd.org/en/publications/2025/07/oecd-employment-outlook-2025_5345f034.html>

OECD

OECD Employment Outlook 2023: Artificial Intelligence and the Labour Market (2023)

The OECD edition that actually covers AI and the labour market, published 11 July 2023. The authority for AI-driven skill-demand shifts. OECD-country focused.

<https://doi.org/10.1787/08785bba-en> <<https://doi.org/10.1787/08785bba-en>>

OECD

Recommendation of the Council on Artificial Intelligence (OECD/LEGAL/0449) (2019)

The OECD AI Principles, the most widely adopted intergovernmental AI governance instrument. Adopted 22 May 2019 and amended 3 May 2024; the 2024 revision is the current text. A recommendation, not legally binding.

<https://legalinstruments.oecd.org/en/instruments/oecd-legal-0449> <<https://legalinstruments.oecd.org/en/instruments/oecd-legal-0449>>

EUROPEAN UNION

Regulation (EU) 2024/1689 (AI Act), as amended by Regulation (EU) 2026/1744 (2024)

Binding law, and the strongest verified legal requirement in this evidence base. Annex III classifies employment and worker-management AI as high-risk; Article 14 mandates human oversight; Article 26 sets deployer obligations. EU jurisdiction. The Digital Omnibus on AI (Regulation (EU) 2026/1744, in force 27 July 2026) delayed application: Annex III high-risk obligations apply from 2 December 2027, Annex I embedded high-risk from 2 August 2028. The duties are enacted but not yet applicable.

<https://ai-act-service-desk.ec.europa.eu/en/ai-act/annex-3> <<https://ai-act-service-desk.ec.europa.eu/en/ai-act/annex-3>>

EUROPEAN COMMISSION

AI Act Service Desk: employment and worker-management use cases (2026)

Commission guidance interpreting how Annex III point 4 applies to workplace systems, including a pay-setting case where human oversight existed formally while the algorithm decided. Guidance, not the Regulation itself, and not independently binding.

<https://ai-act-service-desk.ec.europa.eu/en/ai-act/annex-3> <<https://ai-act-service-desk.ec.europa.eu/en/ai-act/annex-3>>

UNESCO

Recommendation on the Ethics of Artificial Intelligence (2021)

The first global standard-setting instrument on AI ethics, adopted by all member states. Non-binding.

<https://unesdoc.unesco.org/ark:/48223/pf0000381137.locale=en> <<https://unesdoc.unesco.org/ark:/48223/pf0000381137.locale=en>>

INTERNATIONAL TELECOMMUNICATION UNION

Measuring digital development: Facts and Figures 2025 (2025)

The authoritative source for global connectivity and the digital divide. Regional use is reported as a band of 88-93% across CIS, Europe and the Americas, not as a single leading region. ITU revises historical estimates substantially between editions; never compare across editions without the revised series.

<https://www.itu.int/itu-d/reports/statistics/facts-figures-2025/> <<https://www.itu.int/itu-d/reports/statistics/facts-figures-2025/>>

US BUREAU OF LABOR STATISTICS

Employment Projections, 2024-2034 (2025)

The most rigorous official occupational projections available, and the only source here with per-occupation numeric detail. US only and model-based. Fastest growing by percentage and largest absolute growth are different lists.

<https://www.bls.gov/news.release/ecopro.nr0.htm> <<https://www.bls.gov/news.release/ecopro.nr0.htm>>

VACCARO, ALMAATOUQ & MALONE

When combinations of humans and AI are useful: a systematic review and meta-analysis (Nature Human Behaviour 8(12), 2293-2303, DOI 10.1038/s41562-024-02024-1) (2024)

The key peer-reviewed evidence on human-AI collaboration, notable for finding that human-AI combinations often underperform the better of the two alone. Covers studies from January 2020 to June 2023, predating current model generations.

<https://doi.org/10.1038/s41562-024-02024-1> <<https://doi.org/10.1038/s41562-024-02024-1>>

DARON ACEMOGLU

The simple macroeconomics of AI (Economic Policy 40(121)) (2025)

The most rigorous published sceptical estimate of AI's macroeconomic effect, included so the Economy section represents genuine expert disagreement. One model with contested assumptions; others estimate substantially larger effects.

<https://economics.mit.edu/sites/default/files/2024-10/The%20Simple%20Macroeconomics%20of%20AI.pdf> <<https://economics.mit.edu/sites/default/files/2024-10/The%20Simple%20Macroeconomics%20of%20AI.pdf>>

BRYNJOLFSSON, LI & RAYMOND

Generative AI at Work (Quarterly Journal of Economics 140(2), 889-942, DOI 10.1093/qje/qjae044) (2025)

Among the best-identified field evidence on AI's productivity effect on real workers, and the source of the finding that gains concentrate among less experienced workers. Figures here are the published version's, which differ from the withdrawn NBER working paper. One firm, one occupation, one tool, pre-2024 model. Do not extrapolate.

<https://doi.org/10.1093/qje/qjae044> <<https://doi.org/10.1093/qje/qjae044>>

REUTERS

Amazon scraps secret AI recruiting tool that showed bias against women (2018)

The canonical documented case of hiring-algorithm bias. Journalism, not a study: one company, now dated. An illustrative case, never evidence of prevalence.

<https://www.reuters.com/article/world/amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK08J/> <<https://www.reuters.com/article/world/amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK08J/>>

What is not in this list

Figures that could not be verified against the publication itself were left out rather than shown with a hedge, which is why some numbers a reader might expect are absent. The registry behind this report cannot render a figure without a source and a basis, so an unsourced number is not something this interface is able to display. Hover any citation for what its source actually states, including the caveat that limits it. Sources are cited in the form ILO 2025, and the four basis labels are set out at the top of this publication.